



Department of Computer Science

SE4003 – Fundamentals of Software Project Management Spring 2026

Instructor Name: Momna Zaneb

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Office Location/Number:

Office Hours:

TA Name:

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Course Information:

Program: BS (SE) **Credit Hours:** **Course Type:** Core **Prerequisites:** Software Requirements Engineering (SE2001)

Class Meeting Time: Monday - Wednesday 10:00 AM – 11:30 AM

Class Venue: F-311

Course Description/Objectives/Goals:

- Explain principles of project life cycle
- Critically evaluate and discuss the issues around project management and its application in the real world
- Choose project management techniques for IT projects to initiate, plan, execute and evaluate a project and work in teams to create a project plan for a project scenario that includes key tasks, critical path, dependencies and a realistic timeline
- Learn application of tools to facilitate the software project management process (e.g. Microsoft Project).

Course Learning Outcomes (CLOs):

At the end of the course students will be able to: Domain	BT* Level
Understand project management principles and techniques C	1
Apply approaches to manage and optimize the software development processes C	4
Use modern tools to execute and manage the project C	5
* BT= Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A= Affective domain Bloom's taxonomy Levels: 1. Knowledge, 2. Comprehension, 3. Application, 4. Analysis, 5. Synthesis, 6. Evaluation	

Course Textbook:

Software Project Management by Bob Hughes and Mike Cotterell, McGraw-Hill Education; 5th Edition (2009). ISBN-10: 0077122798

Additional references and books related to the course:

Web resources shared on the need basis

Tentative Weekly Schedule

Week	Topic	Contents
1	Introduction	Introduction to software project management. What is a project? How are software projects different from other projects?
2	Process & Methodologies	Contract management and technical project management, activities covered by software project management, plans methods and methodology
3	Project Evaluation	Statement of work, Request for Proposals, Contracts
4	Planning & Estimation	Project planning, resources, return on investment and other evaluation metrics
5		Size Estimation Techniques (Function Points, Use-case Points, etc.)
6		Effort Estimation Techniques (Expert Judgement / WBS, PERT, Parametric models, regression techniques, wideband delphi, cocomo model, user story points estimation etc.)
7		Costing and budgeting, scheduling (Gantt charts / CPM), resource allocation, MS Project tool, JIRA, etc
8	Monitoring & Control	Project progress monitoring, metrics, Presentations on “Personality Assessment Models”
9	Risk Management	Risk analysis, mitigation and management
10	Configuration & Change Management	Configuration items, change control, version control, baselines, change control board
11	People Management	Communication and behavioural issues, team structures, conflict resolution, etc.
12	Process Management	Process definition initiatives, quality standards and frameworks (CMMI, etc.), <u>PMI Process Groups</u>
13	Other Topics	<u>Outsourcing issues, cross-cultural aspects, remote work, etc.</u>
14		Any student presentations, other topics of interest, tools, etc.

(Tentative) Grading Criteria:

Class Participation (5%) Project (15%) Quiz (10 %) Midterms (30 %) Final Exam (40 %)

Course Policies:

- o **Plagiarism** in any work (Quiz, Assignment, Midterms, and Final Exam) from any source, Internet or a Student would result in **F** grade or deduction of absolute marks.
- o 80% attendance is required for appearing in the Final exams.
- o Absolute Grading will be done, inline with department policies.